VolnLabs

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Actryn

controlled execution for AI-directed robots

github.com/volnlabs/actryn

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the problem

the robot moved.
the reply never arrived.

if you resend it

you may repeat a physical action

if you assume it worked

you may hide an incomplete operation

on an inspection round, a timeout alone can’t tell you whether the robot moved. robotics teams can end up building custom authorization, recovery and logging code.

why now

agents can now propose robot actions

AI agent

plans the work and proposes an action

→

the robot side still has to answer

01 is this operation authorized?

02 what limits apply to it?

03 how long does authority last?

04 what if the outcome is uncertain?

→

robot

acts in the physical world, where a rollback can't undo motion

initial focus: inspection robot OEMs and integrators serving energy and industrial sites.

our thesis: better agents, wider robot use, labor constraints and remote inspection all increase the need for governed execution.

the product

Actryn governs the path from agent to robot

01 propose

isolated agent

proposes one typed skill with bounded parameters, not generated code.

→

02 authorize

local operator

grants finite, scoped authority. installing or pairing grants none.

→

03 enforce

Linux authority gate

enforces grants, lifetime and revocation; rejects stale or replayed input.

→

04 reconcile

controller, evidence

reconciles uncertain outcomes before new work and records what is known.

---

one operation identity from proposal to outcome, kept across crashes and restarts

evidence separates admission · dispatch · controller state · observations · missing data

the failure case, handled

Actryn treats a lost reply as unknown

01

send

identity is kept before the request leaves

02

reply lost

the robot may have moved. a timeout can’t say.

03

unknown

`OutcomeUnknown`

no invented success or failure

04

reconcile

inspect the original handle. no automatic resubmission.

05

continue

new work is admitted only after reconciliation

operation in flight

new work blocked until reconciled

new work eligible for authorization

cancellation can target an operation before its first reply arrives, and stale sessions or replayed inputs cannot take authority over replacement work.

what exists today

v0.0.1 runs the full path in software

v0.0.1

software-only developer preview, qualified on 19 october 2026

- ✓ native CLI and Python SDK
- ✓ server and robot services, signed release bundles
- ✓ Linux kernel authority module, run under QEMU
- ✓ cancellation, replay rejection, crash recovery
- ✓ retained operation identity and evidence export

from the release receipt, 19 october 2026

PASS_SOFTWARE_PREVIEW_QUALIFIED

release_qualified	true
independent_installation	false
physical_acceptance	false
live_motion	false

software preview qualified by the founder; not yet a tagged release. independent installation and physical-controller qualification are next.

how we got here

a design partner's feedback shaped Actryn

research · continuing

Axiomos

a standalone programmable robot kernel. it asks how a robot can accept new behavior within explicit limits on authority and physical effects.

design-partner feedback →

they wanted a full driver suite and didn't want to replace their stack for now

product · current focus

Actryn

a narrower product for selected Linux environments that keeps existing drivers and stacks in place. it applies Axiomos' lessons without its kernel.

earlier discovery, recorded 2 october 2026, before the Actryn preview

18 robotics interviews, 10 with OEMs

12 wanted controlled robot execution

5 design-partner conversations

2 pilot-scoping discussions

differentiation

what Actryn adds to existing primitives

failure case	common primitives today	with Actryn
lost reply	ROS 2 actions give each goal an ID and a status; whether to resend is left to the application	the result stays OutcomeUnknown; the original handle is inspected, with no automatic resubmission
crash or restart	the action interfaces define goal states, not recovery after a crash; that is left to the application	unresolved work stays visible across crashes and restarts, and a restart grants no new authority
stale or replayed command	vendor leases such as Spot's reject commands from non-owners or with older ownership	stale generations, replayed inputs and old controller sessions are rejected in the kernel gate
revocation and expiry	leases go stale without keep-alives and can be taken over; E-stops halt motion	finite, scoped grants with set lifetimes and revocation, enforced in the kernel

fleet and inspection platforms such as Korial (formerly Energy Robotics), Formant and Intrinsic work at broader scope. Actryn is a smaller component at the controller boundary and could sit beside them; no integration exists today.

business model

our first target customers build inspection robots

the value we intend to prove

we’re testing whether Actryn reduces the engineering time spent building and maintaining authorization, cancellation and recovery logic for agent-directed robot operations.

pricing hypothesis

US$10k–50k

a one-year pilot or fleet engagement covering integration, qualification and support. not yet validated by sales.

market context

IFR lists 749 professional service-robot producers: a research universe, not a buyer count. bottom-up model: qualified accounts × evidenced contract value, still unvalidated.

revenue path

1. paid pilot or integration
2. reusable controller adapters
3. support and runtime licenses

open source; we plan to charge for integration, qualification and support.

proposed first pilot

one inspection workflow, measured end to end

the setup

buyer: one inspection-robot OEM or integrator

one bounded task, one qualified controller

lab acceptance, with motion observed independently, before field use

metrics Actryn owns

integration hours vs. today's stack

duplicate actuations under fault injection

p99 added command latency

recovery after network or power loss

program outcomes

the site's inspection team benefits. robots can support earlier detection of leaks, corrosion and faults, with fewer site trips. we would record avoided travel and detection-to-response time as program outcomes, not claim them for Actryn.

acceptance limits will come from the partner's workflow and safety requirements.

milestones

each milestone tests a specific risk

0–3 months

install and bench

a second operator installs Actryn; one physical controller is bench-tested

risk tested

single installer, no physical evidence

→

3–6 months

supervised pilot

a supervised design-partner pilot, only after bench acceptance

risk tested

customer value not yet measured

→

6–12 months

repeat and extend

repeat installs and a second controller profile, if reuse holds

risk tested

every robot needs bespoke work

cost should fall as controller adapters, packaging and qualification tests are reused. the measure is how much custom work the second and third installs still need.

team

the founder builds where code meets hardware

Utkarsh Maurya

full-time founder · systems engineer

built Axiomos and Actryn. works on kernels, embedded systems and runtime correctness.

technical mentorship

feedback on Axiomos from engineers in the Cilium and eBPF community

India-based and bootstrapped, not yet incorporated. we plan to add robotics integration expertise as funding allows.

research	two accepted NeurIPS 2026 workshop papers, at AgenticOS and CL4FMAgents
firmware	built Bitcoin-only firmware for Cypherock X1, deployed across 10k+ devices; extended the factory provisioning toolchain
open source	contributions to Limine and bpftime; maintains an rbpf fork
systems	Linux multikernel work; systems and evaluation work at Polymath (YC W26)

what we're looking for

our next goal is one physical deployment that a second team can reproduce and a customer is willing to pay for.

OEMs and integrators

a first pilot: one bounded task on a controller you already run

investors

pre-seed conversations to fund the first physical pilot and an integration hire

test sites and labs

bench access and supervised field time for the first physical tests

expert reviewers

controls, functional-safety and cybersecurity review of the pilot design

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